Test Series No.

1.5

MASTERY TEST SERIES

DURATION: 90 Minutes | **M. MARKS: 400**

DATE: 16/02/2026

SYLLABUS

Physics:	Magnetism and Matter, Electromagnetic Induction, Alternating Current, Electromagnetic Waves, Ray Optics and Optical Instruments + 30% Questions from cumulative syllabus of previously conducted part test
Chemistry:	Alcohols, Phenols and Ethers, Aldehydes, Ketones and Carboxylic Acids, Amines, Biomolecules + 30% Questions from cumulative syllabus of previously conducted part test
Botany:	Molecular Basis of Inheritance, Principles of Inheritance and Variation, Microbes in Human Welfare + 30% Questions from cumulative syllabus of previously conducted part test
Zoology:	Reproductive Health, Evolution, Human Health and Diseases + 30% Questions from cumulative syllabus of previously conducted part test

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **90 minutes** duration and the Test Booklet contains **100** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 400.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. An electromagnetic wave going through vacuum is described by $E = E_0 \sin(kx - \omega t)$. Which of the following is independent of the wavelength?

(1) k (2) $k\omega$

(3) $\frac{k}{\omega}$ (4) both (2) and (3)

2. The magnetic susceptibility has large positive value for:

(1) ferromagnetic material

(2) diamagnetic material

(3) paramagnetic material

(4) Both ferromagnetic and paramagnetic material.

3. A rocket is fired from the Earth towards the Sun. At what approximate distance from the Earth's centre, the gravitational force on the rocket is zero? Mass of the Sun $M_s = 2\times10^{30}$ kg and mass of the Earth $M_e = 6\times10^{24}$ kg. Neglect the effect of other planets etc. (Distance between earth and sun $=1.5\times10^{11}$ m)

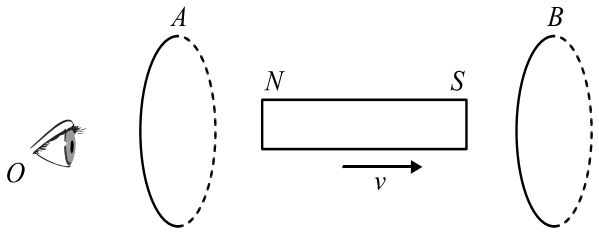
(1) 2.6×10^{8} m

(2) 3.2×10^{8} m

(3) 3.9×10^{9} m

(4) 2.3×10^{9} m

4. A magnet is placed coaxially between two circular conducting loops as shown. As the magnet is moved towards right, the current induced in loops as observed by observer O is;



(1) Zero in both loops

(2) In same direction in both loops

(3) In opposite direction in both loops

(4) Zero in loop A and clockwise in loop B

5. A particle executing S.H.M. has velocities v_1 and v_2 at distance x_1 and x_2 respectively from mean position. The angular velocity (ω) of the particle is given by

(1) $\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$

(2) $\sqrt{\frac{v_2^2 - v_1^2}{x_2^2 - x_1^2}}$

(3) $\sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}}$

(4) $\sqrt{\frac{v_1^2 - v_2^2}{x_1^2 - x_2^2}}$

6. A convex mirror of radius of curvature 1.6m has an object placed in front of mirror at a distance of 1m from it. The image is formed at a distance of-

(1) 6 m behind the mirror

(2) 6 m infront of the mirror

(3) $\frac{4}{9}$m behind the mirror

(4) $\frac{4}{9}$m infront of the mirror

7. For a plane electromagnetic wave propagating along the positive Z-axis, which of the following combinations **correctly** represents the oscillating electric ($\vec{E}$) and magnetic ($\vec{B}$) field components?

(1) $\vec{E}$ along +X-axis, $\vec{B}$ along +Y-axis

(2) $\vec{E}$ along +Y-axis, $\vec{B}$ along +X-axis

(3) $\vec{E}$ along +X-axis, $\vec{B}$ along +Z-axis

(4) $\vec{E}$ along +Z-axis, $\vec{B}$ along +X-axis

8. A student connects a 300 ohm resistor, a 0.9 H inductor, and a 2.0 micro-Farad capacitor in series. An AC source with an angular frequency of 1000 rad/s is applied to the circuit. Find the total impedance of the circuit.

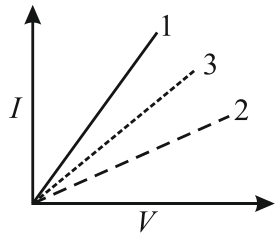
(1) 400 ohms

(2) 500 ohms

(3) 900 ohms

(4) 1300 ohms

9. The graphs between current (I) and voltage (V) for three linear resistors 1, 2 and 3 are given below:



If R_1, R_2 and R_3 are the resistances of these resistors, then which one of the following is correct?

(1) $R_1 > R_2 > R_3$ (2) $R_1 < R_3 < R_2$

(3) $R_3 < R_1 < R_2$ (4) $R_3 > R_2 > R_1$

10. If the angle of minimum deviation of a prism placed in air is equal to its angle of prism, the prism must be made of a material whose refractive index. (prism angle $\leq 90°$)

(1) lies between 1 and $\sqrt{2}$

(2) lies between $\sqrt{2}$ and 2

(3) is less than 1

(4) is greater than 2

11. An air bubble of radius 1 cm rises from the bottom portion through a liquid of density 1.5 g/cc at a constant speed of $0.25\ \text{cm s}^{-1}$. If the density of air is neglected, the coefficient of viscosity of the liquid is approximately, (in Pa-s)

(1) 13000 (2) 1300

(3) 130 (4) 13

12. Match various EM waves in **List-I** with their applications in **List-II**.

List-I (Part of EM Spectrum)		**List-II (Typical Use/ Application)**	
(A)	Microwaves	(I)	TV Remote controls
(B)	Ultraviolet rays	(II)	Medical imaging (bone fracture), security scanners at airport
(C)	Infrared waves	(III)	Radar systems, microwave ovens
(D)	X-rays	(IV)	Sterilization of medical equipment, water purification

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II

(2) A-III, B-I, C-IV, D-II

(3) A-II, B-IV, C-III, D-I

(4) A-I, B-II, C-III, D-IV

13. A bar magnet placed in a uniform magnetic field has:

(1) maximum potential energy when the torque on it is maximum.

(2) zero potential energy when the torque on it is maximum.

(3) zero potential energy when the torque on it is zero.

(4) minimum potential energy when the torque on it is maximum.

14. An alternating current (AC) flows through a wire with a frequency of 50 Hz and an RMS value of 10 Amperes. Calculate the ratio of the Peak Current to the RMS Current for this sinusoidal wave.

(1) 1 (2) 1/2

(3) $\sqrt{2}$ (4) $1/\sqrt{2}$

15. A disc of radius 0.4 m and mass 1 kg rotates about an axis passing through its centre and perpendicular to its plane. The angular acceleration is 10 rad/s^2. The tangential force applied to the rim of the disc is

(1) 2 N (2) 4 N

(3) 3 N (4) 1 N

16. The refractive index of a converging lens is 2. What will be the power of this lens if it is placed in a medium of same refractive index? (Assume the radii of curvature of faces of lens are R_1 and R_2 respectively)

(1) Zero

(2) Infinite

(3) 1

(4) $\dfrac{R_1 R_2}{R_1 - R_2}$

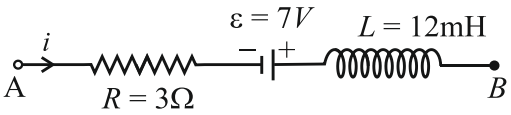
17. A straight conductor of length 50 cm is moving perpendicular to a uniform magnetic field with a speed of 72 km/h. If the length of conductor, its velocity and magnetic field are mutually perpendicular, induced EMF developed across the ends of the conductor is 10 V, calculate the magnitude of the magnetic field.

(1) 0.5 T (2) 1.0 T

(3) 2.0 T (4) 5.0 T

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18. The electric network shown in figure is part of a complete circuit. If at any instant, current $i = 2A$ and potential difference across A and B is zero, then value of $\left|\frac{di}{dt}\right|$ is:



(1) $\frac{26}{3}$ A/s (2) $\frac{25}{3}$ A/s

(3) $\frac{250}{3}$ A/s (4) $\frac{290}{3}$ A/s

19. A ray of light from air strikes a transparent rectangular slab of thickness 20 cm and refractive index $\sqrt{2}$ at an angle of incidence 60° and finally emerge to the air. The angle between the incident and emergent ray will be:

(1) 90°
(2) 45°
(3) 60°
(4) Zero

20. An electric dipole is placed in a uniform electric field with the dipole moment making an angle θ with the electric field. The value of θ for stable equilibrium is

(1) $\frac{\pi}{6}$ rad

(2) $\frac{\pi}{3}$ rad

(3) 0 rad

(4) $\frac{\pi}{2}$ rad

21. A block of mass 8 kg is suspended by a vertical rope of length 3m from the ceiling. A force of 40 N is applied horizontally to the block. Then the angle that the rope makes with the vertical in equilibrium is (acceleration due to gravity $= 10\ ms^{-2}$, neglect the mass of the rope)

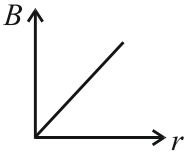
(1) $\sin^{-1}\left(\frac{1}{2}\right)$

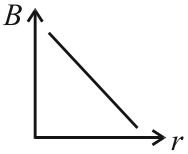
(2) $\tan^{-1}\left(\frac{1}{2}\right)$

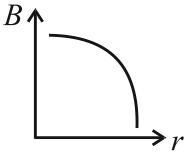
(3) $\sin^{-1}\left(\frac{1}{3}\right)$

(4) $\tan^{-1}\left(\frac{1}{3}\right)$

22. Which of the following graphs shows the variation of magnetic field B with perpendicular distance r from a straight current carrying thin long wire?

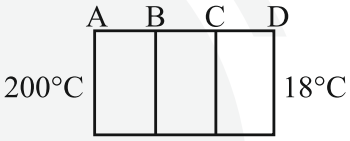
(1) 

(2) 

(3) 

(4) 

23. Three bars AB, BC and CD of equal lengths and equal area of cross section are connected in series. Their thermal conductivities are in the ratio K_{AB} : K_{BC} : K_{CD} = 2:4:3. If the open ends of the first and last bar are at temperature 200°C and 18°C respectively, then in the steady state the temperature of the 1st Junction (B) will be



(1) 96° C
(2) 116° C
(3) 129° C
(4) 132° C

24. A metallic rod of length 2m is rotated with frequency of $\frac{50}{\pi}$ revolution per second with one end hinged. A constant magnetic field of 0.5 T, parallel to axis of rotation is present. The emf induced between ends of the rod is:

(1) 25 V
(2) 200 V
(3) 50 V
(4) 100 V

25. In an electromagnetic wave propagating in free space, the root mean square value of the electric field is 6V m^{-1}. The peak value of the magnetic field is:

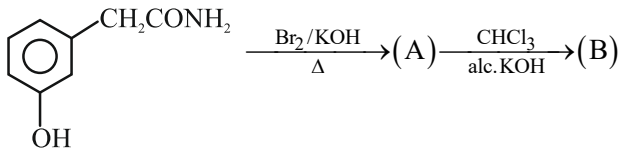
(1) $2\sqrt{2}\times10^{-8}$ T

(2) $3\sqrt{2}\times10^{-8}$ T

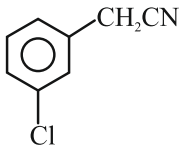
(3) $4\sqrt{2}\times10^{-8}$ T

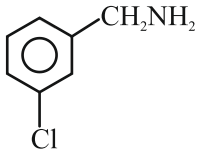
(4) $5\sqrt{2}\times10^{-8}$ T

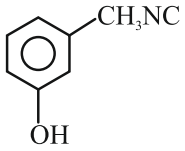
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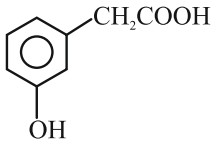
26. 

The final product (B) is:

(1) 

(2) 

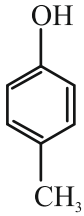
(3) 

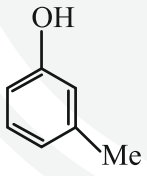
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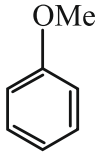
27. An aqueous solution of a compound AB freezes at –0.48°C. At this temperature, AB behaves as a non-electrolyte. The same solution boils at 100.26°C at a P of 1 atm (for water, $\frac{K_f}{K_b} = 3.7$) we can conclude that compound AB at the boiling point of solution,

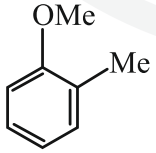
(1) Behaves as a non-electrolyte
(2) Ionises to the extent of 50%
(3) Dimerises completely in solution
(4) Dissociation in solution

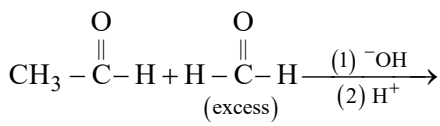
28. Compound (X), C_7H_8O is soluble in NaOH but not in $NaHCO_3$. (X) reacts rapidly with Br_2 to give (Y), $C_7H_5OBr_3$. Probable structure of X would be:

(1) 

(2) 

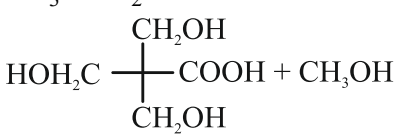
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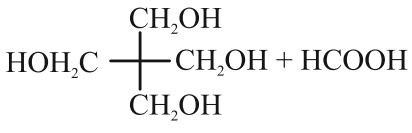
(4) 

29.  . The major product which can be isolated from this reaction is:

(1) $CH_3COOH + CH_3OH$

(2) $CH_3 - CH_2OH + HCOOH$

(3) 

(4) 

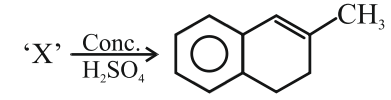
30. In the given statements;

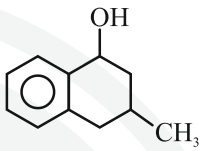
(A) Glucose is a monosaccharide
(B) Sucrose is a disaccharide
(C) Maltose is a monosaccharide

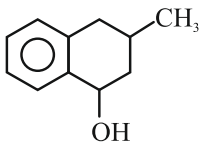
Choose the **correct** answer from the options given below:

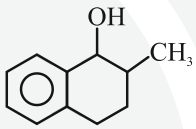
(1) A & B only
(2) A & C only
(3) A only
(4) B & C only

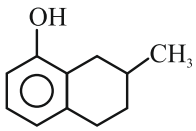
31. In the given reaction, the compound X is:



(1) 

(2) 

(3) 

(4) 

32. Given below are two statements:

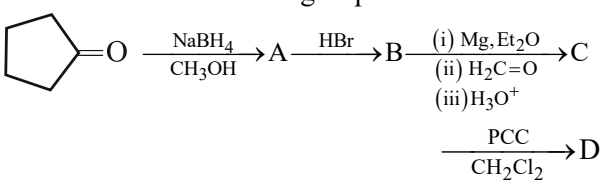
Statement I: Kohlrausch law helps to find the molar conductivity of weak electrolyte at infinite dilution.

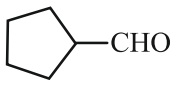
Statement II: Conductivity decreases but molar conductivity increases with decrease in concentration.

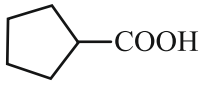
In the light of the above statements, choose the *most appropriate* answer from the options given below:

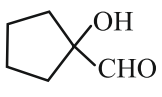
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

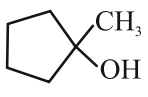
33. What is D in the following sequence of reactions?



(1) 

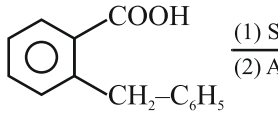
(2) 

(3) 

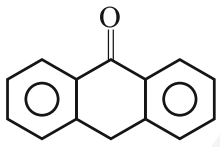
(4) 

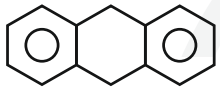
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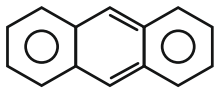
34. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Phenols give o– and p-nitrophenol on nitration with conc. HNO_3 and H_2SO_4 mixture.
Reason R: —OH group in phenol is o-, p-directing.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

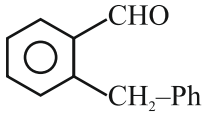
35. 

B is:

(1) 

(2) 

(3) 

(4) 

36. Two substances A and B are present such that $[A_0] = 4[B_0]$ and half-life of A is 5 minute and that of B is 15 minute. If they start decaying at the same time of following first order kinetics, how much time later will the concentration of both of them would be same?
(1) 15 minute (2) 10 minute
(3) 5 minute (4) 12 minute

37. Match the item given in **List-I** with items given **List-II.**

List-I		List-II	
(A)	Lysine	(I)	Neutral optically active amino acid
(B)	Aspartic acid	(II)	Basic amino acid
(C)	Valine	(III)	Neutral optically inactive amino acid
(D)	Glycine	(IV)	Acidic amino acid

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-IV, C-I, D-III

38. Given below are two statements:
Statement I: The reaction between a diazonium salt and an aromatic amine or a phenol, giving an amino azo or hydroxy azo compound is called coupling reaction.
Statement II: Condensation of diazonium salt with phenol is carried out in weakly acidic medium.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

39. Ether is obtained from ethyl alcohol in presence of H_2SO_4 at:
(1) 113 K
(2) 443 K
(3) 413 K
(4) 213 K

40. Which of them is **correct** order for solvolysis rate in aqueous acetone? (via S_N1)
P. $CH_3O - CH = CH - CH_2 - Cl$
Q. $CH_2 - O - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2 - Cl$
R. $CH_3 - CH_2 - CH = CH - Cl$
S. $CH_3O - CH_2 - CH_2 - CH_2 - Cl$
(1) R > P > Q > S
(2) P > Q > R > S
(3) Q > P > S > R
(4) P > Q > S > R

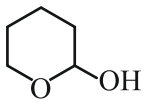
41. In presence of a catalyst, the activation energy is lowered by 2 kcal at 127°C. The rate as compared to old rate is:
(1) e^2 times
(2) e^{-2} times
(3) $e^{2.5}$ times
(4) $e^{-4.5}$ times

42. The major product of the reaction between tert – butyl chloride and sodium ethoxide is:
(1) 2-methyl prop-1-ene
(2) 1-butane
(3) 2-butane
(4) Ethene

[Contd...

43. In the following compounds

(A) $CH_3-\overset{\overset{O}{\|}}{C}-H$

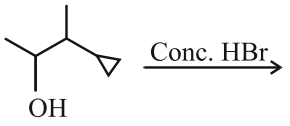
(B) 

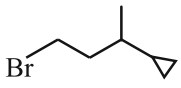
(C) $CH_3-CH_2-CH(OH)(OCH_3)$

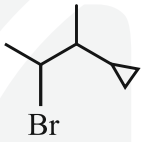
(D) $PhCOCH_3$

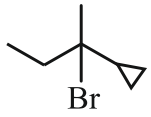
Which of the compounds will give positive tollen's text

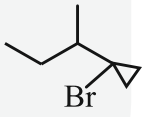
(1) A, B & D only (2) A, B & C only
(3) B, C & D only (4) A, C & D only

44.  Product is:

(1) 

(2) 

(3) 

(4) 

45. For $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$, $E^o = +1.33V$ at $\left[Cr_2O_7^{2-}\right] = 4.5$ millimole per Litre, $\left[Cr^{3+}\right]$ = 15 millimole per Litre, E is 1.06 V at 298 K. The pH of the solution is nearly equal to:

(1) 2 (2) 3
(3) 5 (4) 4

46. Which of the following statements regarding chemical properties of acetophenone are **wrong**?

I. It is reduced to methyphenylcarbinol by sodium and ethanol
II. It is oxidized to benzoic acid with acidified $KMnO_4$.
III. It does not undergo aldol condensation.
IV. It does not undergo iodoform reaction with iodine and alkali

(1) I and II (2) II and IV
(3) III and IV (4) I and III

47. At 500 K, the half life period of a gaseous reaction at an initial pressure of 80 Pa is 350 sec. When the pressure is 40 Pa, the half life period is 175 sec. The order of the reaction is:

(1) zero (2) one
(3) two (4) three

48. Select the **incorrect** statement.

(1) Benzenediazonium chloride is prepared by the reaction of aniline with nitrous acid at 273-278 K.
(2) Benzene diazonium chloride is a colourless crystalline solid.
(3) Benzenediazonium chloride is readily soluble in water and is stable in cold but react with water when warmed.
(4) Benzenediazonium fluoroborate is water soluble and is unstable at room temperature.

49. Match the reactions given in **List-I** with the suitable reagents given in **List-II.**

	List-I (Reactions)		**List-II (Reagents)**
(A)	Benzophenone→ Diphenylmethane	(I)	$LiAlH_4$
(B)	Benzaldehyde→ 1-Phenylethanol	(II)	DIBAL-H
(C)	Cyclohexanone→ Cyclohexanol	(III)	Zn(Hg)/conc. HCl
(D)	Phenyl benzoate Benzaldehyde	(IV)	CH_3MgBr

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-I, C-III, D-II
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-III, C-II, D-I

50. Which of the following statement(s) regarding preparation of haloalkanes is/are correct?

(a) The reactions of primary and secondary alcohols with HCl require the presence of a catalyst, $ZnCl_2$.
(b) The order of reactivity of alcohols with a given haloacid is 3°>2°>1°.
(c) In Finkelstein reaction, alkyl chlorides are often prepared by the reaction of alkyl iodides/ bromides with $ZnCl_2$, in dry acetone.
(d) The synthesis of alkyl fluorides is best accomplished by heating an alkyl chloride/bromide in the presence of a metallic fluoride such as AgF, Hg_2F_2, CoF_2, or SbF_3.

(1) (a) and (b) only
(2) (b), (c) and (d) only
(3) (a), (b) and (d) only
(4) (b) and (c) only

[Contd...

51. The 'ears' in a corn cob are:
(1) stigma and style.
(2) filament and anthers.
(3) calyx and corolla.
(4) petals and thalamus.

52. Match **List-I** with **List-II**.

List-I (Genes of *lac* operon)		**List-II (Functions)**	
(A)	Structural gene	(I)	Provides attachment site for RNA polymerase
(B)	Regulator gene	(II)	Controls the activity of the operator gene
(C)	Promoter	(III)	Interacts with repressor protein
(D)	Operator	(IV)	Transcribes *lac* mRNA

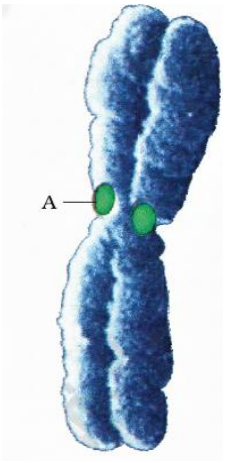
Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-I, D-II
(2) A-III, B-IV, C-II, D-I
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-II, C-I, D-III

53. Trisomy of a sex chromosome is observed in:
(1) Turner's syndrome
(2) Myotonic dystrophy
(3) Klinefelter's syndrome
(4) Down's syndrome

54. Lactic Acid Bacteria (LAB):
(1) increase the vitamin d content of curd.
(2) check the growth of disease-causing microbes in gut.
(3) produce acids that completely digest milk proteins.
(4) are used commercially for the production of butyric acid.

55. Some steps involved in the classical technique of DNA fingerprinting are listed below. Identify the option that shows the correct sequence of these steps.
A. Hybridisation
B. Restriction digestion using endonucleases
C. Autoradiography
D. Gel electrophoresis
E. Blotting
F. Isolation of DNA
Choose the **correct** sequence:
(1) F → B → E → D → A → C
(2) D → B → E → C → A → F
(3) F → B → D → E → A → C
(4) D → B → A → E → C → F

56. Identify the **correct** statement about 'A'.



(1) It is located on the surface of the secondary constriction.
(2) It serves as the site of attachment of spindle fibers to the chromosomes.
(3) It is a histone protein that helps in packaging.
(4) It is a satellite.

57. Histone proteins are rich in:
(1) phenylalanine and arginine.
(2) lysine and arginine.
(3) serine and arginine.
(4) tyrosine and arginine.

58. Given below are two statements.
Statement I: In photophosphorylation, it is the light energy that is utilised for the production of proton gradient required for phosphorylation.
Statement II: In respiration, it is the energy of oxidation-reduction utilised for the production of proton gradient required for phosphorylation.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

59. In which of the following organisms do half of the sperm produced carry only one sex chromosome along with autosomes, while the other half carry only autosomes?
(1) Hen
(2) *Drosophila*
(3) Grasshopper
(4) Honey bee

[Contd...

60. Which of the following best describes the approach of an organic farmer towards pests in agriculture?
(1) All pests are kept as it is to make vibrant natural ecosystem.
(2) Pests should be controlled through the use of chemical pesticides.
(3) Pests should be kept at manageable levels, as they help maintain the balance of the ecosystem.
(4) Pests should be fully eradicated, as they are always harmful to ecosystem.

61. Given below are two statements.
Statement I: On the DNA strand that acts as the template with 5′ → 3′ polarity, replication is continuous.
Statement II: DNA strand that acts as the template with 3′ → 5′ polarity, replication is discontinuous.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

62. Which of the following is **not** a characteristic feature of monocot plants?
(1) The leaf base expands into a sheath, partially or wholly covering the stem.
(2) The stem possesses water-containing cavities and phloem parenchyma.
(3) The root shows a polyarch condition.
(4) Monocots generally have endospermic seeds.

63. In *Drosophila*, the genes for eye colour and wing type:
(1) Show a recombination frequency of 62.8%
(2) Are very tightly linked and show more parental types in the offspring
(3) Are loosely linked and show higher recombination
(4) Show only 1.3% recombination

64. Which hormone promotes root growth and root hair formation, thus helping the plants to increase their absorption surface?
(1) Gibberellins
(2) Zeatin
(3) ABA
(4) Ethylene

65. In a typical nucleus, some regions of chromatin are loosely packed and are:
(1) Referred to as heterochromatin.
(2) More densely packed.
(3) Said to be transcriptionally active chromatin.
(4) Organised into nucleoids in large loops held by proteins.

66. Match **List-I** with **List-II**.

List-I (Microorganism)		**List-II (Product/Bioactive molecule)**	
(A)	*Aspergillus niger*	(I)	Statins
(B)	*Propionibacterium sharmanii*	(II)	Cyclosporin A
(C)	*Monascus purpureus*	(III)	Swiss cheese
(D)	*Trichoderma polysporum*	(IV)	Citric acid

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-I, D-II
(3) A-IV, B-II, C-I, D-III
(4) A-III, B-I, C-IV, D-II

67. All of the following statements are correct about α-thalassemia, **except**:
(1) It is controlled by a single HBB gene located on chromosome 11 of each parent.
(2) It involves the formation of abnormal haemoglobin molecules.
(3) It leads to anaemia, which is characteristic of the disease.
(4) It occurs due to mutation or deletion of one or more of the four α-globin genes.

68. Anaphase I differs from anaphase II in that:
(1) the former involves separation of homologous chromosomes, whereas the latter involves separation of sister chromatids.
(2) the former involves compaction of chromosomes, whereas the latter involves decondensation of chromosomes.
(3) the former involves disintegration of the nuclear membrane, whereas the latter involves reappearance of the nuclear envelope.
(4) the former involves pairing of homologous chromosomes, whereas the latter involves separation of homologous chromosomes.

[Contd...

69. A molecule that can act as genetic material must fulfill the following criteria, **except** that it:

(1) should be able to express itself in the form of mendelian characters.

(2) should be able to generate its replica.

(3) should provide scope for fast changes.

(4) should be chemically and structurally stable.

70. Mendel selected as pairs those pea plant varieties which were similar in all traits except for one contrasting character. How many true-breeding pea plant varieties did he use?

(1) 14 true-breeding pea plant varieties

(2) 7 true-breeding pea plant varieties

(3) 2 true-breeding pea plant varieties

(4) 4 true-breeding pea plant varieties

71. Which of the following is **incorrect** statement?

(1) DNA fingerprinting forms the basis of paternity testing in case of disputes.

(2) Depending on base composition (A:T-rich or G:C-rich), length of the segment, and number of repetitive units, satellite DNA is classified into microsatellites and minisatellites.

(3) Polymorphisms are non-inheritable from parents to children.

(4) The technique of DNA fingerprinting was initially developed by Alec Jeffreys.

72. Which of the following statements is **correct** about phenylketonuria?

(1) There is an increase in tyrosine level in the affected individual.

(2) There is a lack of the enzyme phenylalanine hydroxylase.

(3) It is a sex-linked recessive trait.

(4) It is a type of chromosomal genetic disorder.

73. Identify the **incorrectly** matched pair with respect to a pest/pathogen and its biocontrol agent.

(1) Butterfly caterpillar – *Bacillus thuringiensis*

(2) Several plant pathogens – *Trichoderma* species

(3) Mosquitoes – Red and black markings beetle

(4) Aphids – Ladybird

74. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Viruses did not find a place in the five kingdom classification of Whittaker's classification.

Reason R: Viruses are not considered truly 'living', as they are non-cellular organisms that are characterised by having an inert crystalline structure outside the living cell.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

75. Identify the incorrectly matched pair with respect to the plant group and its example.

(1) Gymnosperm – *Cedrus*

(2) Angiosperm – *Salvia*

(3) Pteridophyte – *Adiantum*

(4) Bryophyte – *Fucus*

76. Which group of hormones interacts with intracellular receptors and regulates gene expression?
(1) Peptide and Protein hormones
(2) Amino-acid derivatives like Epinephrine
(3) Steroids and Iodothyronines
(4) Pituitary and Hypothalamic hormones

77. Consider the following statements about neurons and nerve fibres:
I. Synaptic knobs contain synaptic vesicles filled with neurotransmitters.
II. Axons conduct nerve impulses towards the cell body.
III. Multipolar neurons have one axon and two or more dendrites and are commonly found in the cerebral cortex.
IV. Nodes of Ranvier are gaps between two adjacent myelin sheaths in myelinated nerve fibres.
V. Unmyelinated nerve fibres lack Schwann cells.
Which of the above statements are **correct**?
(1) I, II and IV only
(2) I, III and IV only
(3) I, III, IV and V only
(4) I, II, IV and V only

78. Given below are two statements.
Statement I: The foetus receives some antibodies from the mother through the placenta during pregnancy, which is an example of passive immunity.
Statement II: The yellowish fluid colostrum secreted by a mother during the initial days of lactation contains abundant IgE antibodies to protect the infant.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

79. Saheli, the oral contraceptive for females, is unique because:
(1) it is a steroidal preparation taken daily.
(2) it is a 'once a week' pill with very few side effects.
(3) it was developed by WHO scientists in Geneva.
(4) it must be taken within 72 hours of coitus to be effective.

80. Match **List-I** with **List-II**.

List-I		List-II	
(A)	350 mya	(I)	Fish with stout/strong fins (Lobefins)
(B)	320 mya	(II)	First cellular forms appeared
(C)	65 mya	(III)	Sea weeds and few plants existed
(D)	2000 mya	(IV)	Dinosaurs suddenly disappeared

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-II, B-I, C-III, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-I, B-III, C-IV, D-II

81. During the life cycle of *Plasmodium*, what is the immediate cause of the recurring chills and high fever observed every three to four days in the host?
(1) Immediately after the entry of sporozoites into the human body via the mosquito bite.
(2) The multiplication of parasites within the liver cells.
(3) The rupture of red blood cells and the subsequent release of haemozoin.
(4) The development of sexual stages (gametocytes) in the red blood cells.

82. Identify the **correct** statements about sterilization methods:
A. They are generally advised as a terminal method to prevent further pregnancies.
B. In vasectomy, a small part of the fallopian tube is removed or tied up.
C. These techniques are highly effective but their reversibility is very poor.
D. Surgical intervention blocks gamete transport.
(1) A and B only (2) B and C only
(3) A, C and D only (4) A, B, C and D

[Contd...

83. Which of the following is **correct** regarding the structure of a human sperm?
(1) The middle piece contains numerous mitochondria which produce energy for the movement of the head.
(2) The acrosome is filled with enzymes that help in the nourishment of the sperm.
(3) The sperm head contains an elongated haploid nucleus covered by the acrosome.
(4) For normal fertility, at least 40 percent of the total ejaculated sperms must show normal shape and size.

84. What did the Big Bang theory attempt to explain?
(1) The origin of the first self-replicating metabolic capsule.
(2) The singular huge explosion that led to the expansion of the universe.
(3) The transfer of 'spores' from outer space to Earth.
(4) The chemical evolution of proteins and RNA from inorganic molecules.

85. Given below are two statements.
Statement I: A single U-shaped bone called hyoid is present at the base of the buccal cavity and is a part of facial bone.
Statement II: Sliding filament theory states that contraction of a muscle fibre takes place by the sliding of the thin filaments over the thick filaments.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

86. Regarding Assisted Reproductive Technologies (ART), which of the following statements are **correct**?
A. ZIFT involves the transfer of the zygote or early embryos (up to 8 blastomeres) into the fallopian tube.
B. GIFT involves the transfer of an embryo with more than 8 blastomeres into the uterus.
C. ICSI is a procedure where a sperm is directly injected into the ovum.
D. IUI is a technique where semen is artificially introduced into the uterus.
(1) A and B only
(2) A, B and D only
(3) B and D only
(4) A, C and D only

87. Identify the **correct** sequence of human evolution based on brain capacity (from lowest to highest).
(1) *Homo erectus* (650-800cc) → *Homo habilis* (900cc) → Neanderthal man (1400cc).
(2) *Homo habilis* (650-800cc) → *Homo erectus* (900cc) → Neanderthal man (1400cc).
(3) *Australopithecines* (900cc) → *Homo habilis* (1400cc) → *Homo sapiens* (650-800cc).
(4) *Homo habilis* (900cc) → *Homo erectus* (650-800cc) → Neanderthal man (1400cc).

88. Regarding the replication of HIV in the human body, which of the following statements are **correct**?
A. Viral RNA is converted to viral DNA by the enzyme reverse transcriptase.
B. The virus first enters the T-lymphocytes and then the macrophages.
C. Macrophages act like an 'HIV factory' by continuing to produce virus particles.
D. A progressive increase in helper T-lymphocytes occurs during the infection.
(1) A and B only
(2) A and C only
(3) B and D only
(4) A, C and D only

89. Amniocentesis is a procedure used for _____, but is currently under a statutory ban for _____.
Choose the **correct** option for filling the blanks.
(1) determining fetal age; curing Down syndrome.
(2) testing for genetic disorders; sex-determination to check female foeticide.
(3) inducing MTP; detecting STIs.
(4) measuring amniotic fluid volume; checking for twins.

90. What did Karl Ernst von Baer note that led to the disapproval of Ernst Haeckel's embryological support for evolution?
(1) Embryos of all vertebrates develop vestigial gill slits.
(2) Embryos always pass through the adult stages of other animals.
(3) Embryos never pass through the adult stages of other animals.
(4) Human adults possess functional gills like fish.

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91. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Hydrolases	(I)	Enzymes that catalyse removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds
(B)	Lyases	(II)	Enzymes catalysing hydrolysis of ester, ether, peptide, glycosidic, C-C, C-halide or P-N bonds
(C)	Isomerases	(III)	Includes all enzymes catalysing inter-conversion of optical, geometric or positional isomers
(D)	Ligases	(IV)	Enzymes catalysing the linking together of 2 compounds

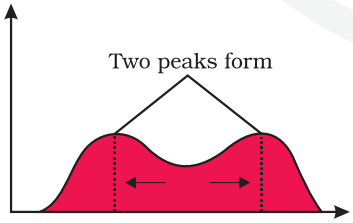
Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-II, D-I
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-I, C-III, D-IV

92. Which of the following properties distinguishes a malignant tumor from a benign tumor?

(1) It remains confined to its original location and causes little damage.
(2) It shows the property of contact inhibition.
(3) It exhibits metastasis, reaching distant sites through blood to start new tumors.
(4) It is a mass of cells that has stopped dividing.

93. Which of the following options **best** exemplify the type of natural selection represented by the diagram given below?



(1) In a population of snails, both very light and very dark shells are favored in patchy habitats, while medium-colored shells are preyed upon more.
(2) In human infants, intermediate birth weight has the lowest mortality, while extremely low or high weights face higher risks.
(3) Moths in industrial England showed increased frequency of dark-colored forms over time and showed decreased frequency of white-colored forms.
(4) In a forest, only deer with average running speed survive, while the slowest and fastest are predated or isolated.

94. Match **List-I** with **List-II**.

List-I (Contraceptive method)		List-II (Mechanism)	
(A)	IUDs	(I)	Prevents conception by blocking entry of sperm through cervix
(B)	Barrier (Diaphragm)	(II)	Increase phagocytosis of sperm within uterus
(C)	Natural (Withdrawal)	(III)	Blocks gamete transport
(D)	Surgical	(IV)	Male withdraws penis just before ejaculation

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-IV, D-III
(2) A-II, B-III, C-IV, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-IV, B-I, C-II, D-III

95. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The counter current mechanism in the kidney helps to maintain an increasing osmolarity gradient from the cortex to the inner medulla.

Reason R: The special arrangement of Henle's loop and *vasa recta* allows the exchange and recycling of NaCl and ammonia, which maintains the medullary osmotic gradient.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

96. Which of the following statements **correctly** explains the clinical significance of an ECG?

(1) The ECG records only atrial activity and is mainly used to study atrial disorders.
(2) The number of P-waves per minute determines the heart rate of an individual.
(3) Any deviation from the normal ECG wave pattern may indicate a cardiac abnormality.
(4) The T-wave represents the beginning of ventricular systole.

97. Which of the following are examples of adaptive radiation?

A. Darwin's Finches in Galapagos Islands.
B. Australian marsupials evolving from an ancestral stock.
C. The evolution of the thorns in *Bougainvillea* and tendrils of *Cucurbita*.
D. Placental mammals in Australia evolving into varieties of such placental mammals each of which appears to be 'similar' to a corresponding marsupial.

(1) A and B only
(2) A, B, and D only
(3) B and C only
(4) A, B, C and D

98. Given below are two statements.

Statement I: MTPs are considered relatively safe during the second trimester.

Statement II: In India, MTP was legalized in 1951 along with the Family Planning programme.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

99. Identify how many statements are **correct** regarding drugs given below.

A. Heroin is chemically diacetylmorphine and is a stimulant.
B. *Cannabinoids* are obtained from the inflorescences of *Cannabis sativa*.
C. Cocaine is obtained from *Erythroxylum coca* and interferes with dopamine transport.
D. Morphine is extracted from the latex of the *Papaver somniferum* plant.

(1) One
(2) Two
(3) Three
(4) Four

100. Which component of the immune system is primarily responsible for the graft rejection?

(1) Humoral immune response mediated by B-lymphocytes.
(2) Cell-mediated immune response mediated by T-lymphocytes.
(3) Physiological barriers like acid in the stomach and tears.
(4) Innate immunity present at the time of birth.

■■■

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